THE UK UNIVERSITY INTEGRATION BEE 2025/2026

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ROUND TWO

CROSSNUMBER ROUND

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ACROSS

3. A four-digit number $\overline{abcd}$ where $\overline{ab} = \left\lceil \int_2^5 f(x)\,\mathrm{d}x \right\rceil$ and $\overline{cd} = \left\lceil \int_2^6 f(x)\,\mathrm{d}x \right\rceil$ for a certain non-negative function f on $[2,6]$. This function f also satisfies the following properties:
   - $\int_5^6 f(x)\,\mathrm{d}x$ is a perfect number (i.e. a positive integer equal to the sum of all its proper divisors).
   - $\left(\int_2^5 f(x)\,\mathrm{d}x\right)^2 + \left(\int_2^6 f(x)\,\mathrm{d}x\right)^2 = 1076$.

5. A four-digit number $\overline{abcd}$ such that
   - $\int_0^1 \frac{(1+2e^a x+e^{2a})\ln(x)}{x+e^a+e^{2a}x+e^a x^2}\,\mathrm{d}x = \frac{4004-2\times(3\text{ ACROSS})-\pi^2}{b-3}$, and
   - $\overline{cd}$ divides at least one of the answers in this crossnumber.

7. The answer to this clue satisfies the following two conditions:
   - $(7\text{ ACROSS}) \equiv 1 \pmod{10}$.
   - $\left\lceil \frac{(7\text{ ACROSS})}{10} \right\rceil \left| \int_1^\infty \frac{\sin(\pi x)}{\lfloor x \rfloor^2}\,\mathrm{d}x \right| = 2\left\lfloor \sqrt{(8\text{ DOWN})} \right\rfloor \pi$.

9. The middle digit is the arithmetic mean of the digits of (4 DOWN).

11. $(16\text{ DOWN}) \int_{-\pi}^{\pi} \frac{|\sin(x)|}{1+(2+5x^2)^x}\,\mathrm{d}x + (24\text{ ACROSS}) \int_0^\pi \ln\left(\sqrt{\frac{(1-\cos(x))^{\cos(x)}}{(1-\sin(x))^{\sin(x)}}}\right)\mathrm{d}x$.

12. A palindromic number.

15. A square number whose first and last digits are the same.

22. (20 DOWN) is the arithmetic mean of (7 ACROSS) and this number.

24. The value of $f\left(e^{56}\right)$, where $f(z) = \lim_{y\to\infty} \int_e^z \int_0^y y t^{x-2} e^{t^x - y^x}\,\mathrm{d}t\,\mathrm{d}x$.

In the answers to the following clues, nine of the digits 0-9 appear once and exactly once, and one digit does not appear at all. The digit that doesn't appear is the first digit of (9 ACROSS).

17. Each digit is a power of 2.

21. A three-digit number $\overline{abc}$ with all digits divisible by 3 such that 9 divides $b+c$ and $c > 0$.

23. The first and last digits are happy, while the middle digit is unhappy. Here's more information:
   - The middle digit is a prime that doesn't divide (21 ACROSS).
   - The last digit makes $\gcd\left((19\text{ DOWN}), \left\lfloor \frac{(12\text{ ACROSS})}{10} \right\rfloor\right)$ bigger than 3.